Platform Constraints: The primary deliverable **must be a native/cross-platform mobile application** (built with Flutter, React Native, .NET MAUI, Kotlin, or Swift). Web or desktop apps are only allowed for the standalone Administrator Panel.

Comprehensive Project Brief & Feature Requirements

User Types & Role-Based Access (RBAC)

Must enforce strict backend role-based access control across four distinct roles:

- **Students:** Enrolled students registered using official @my.richfield.ac.za.
- **Alumni:** Graduates requiring a custom, researched identity verification flow (since institutional emails may be inactive).
- **Business Users:** Verified recruiters/employers requiring manual administrative approval before accessing the platform.
- **Administrators:** Richfield staff with secure, non-self-registerable accounts to oversee content, users, and events.

Core Platform Modules

- **Digital Portfolio Profiles:** Extensive profiles including bio, skills, GitHub repositories, live application links, Credly badges, academic achievements, CV uploads, and skill endorsements/recommendations.
- **Social & Networking Feed:** Personalised activity feed (posts, videos, day-in-the-life stories), connection requests, direct messaging, and post engagement.
- **Career & Opportunity Features:** Business job postings (internships/vacancies), student application tracking, institutional event listings, and an **Alumni Career Pathway Explorer**.
- **Data Analytics Dashboards:** Three visually distinct dashboards using charts/graphs tailored for **Students** (views, skill demand), **Business Users** (applicant pipeline), and **Administrators** (platform usage, flags).

Mandatory AI & Advanced Technical Features

- **AI Profile Assistant & Chatbot:** Context-aware onboarding guide to help students build compelling profiles, suggest improvements, and answer platform queries.
- **NLP-Assisted Profile Building:** Automated skill and qualification extraction when users upload a CV or enter bio text.
- **Smart Job Matching:** Automated recommendation engine matching new opportunities to student profiles based on skills and programme alignment.

- **Real-Time Notifications & Media Processing:** Live push notifications via WebSockets/push services and backend video compression/thumbnail generation for feed uploads.